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Palaeomagnetic rotations in the eastern Betic Cordillera, southern Spain

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ABSTRACT

Palaeomagnetic declinations from the External Zones of the eastern Betic Cordillera (southern Spain) and an adjacent area of the Internal Zone indicate variable and locally very large clockwise rotations. The rotations occurred after latest Oligocene times, and probably before the late Miocene. This overlaps a period of dextrally oblique convergence during the early to middle Miocene along the Internal/External Zone Boundary and within the External Zone.

Rotations in the External Zone are recorded by palaeomagnetic results from 27 sites in Upper Jurassic (*Ammonitico rosso* facies) limestones from the Subbetic Zone, and one site of a similar age from the Prebetic. The sites show either two- or three-component behaviour. The low-temperature component is coincident with the present-field direction. The intermediate-temperature component was probably acquired during Miocene folding. The high-temperature component passes fold tests (Miocene age) and a conglomerate test (Eocene–early Oligocene age).

Palaeomagnetic declinations from individual tectonic blocks in the Subbetic are consistent, but indicate large differential rotations between blocks. These blocks are underlain by low-angle thrust faults, which probably accommodated much of the rotation. The largest rotations occur on relatively small isolated blocks of Jurassic carbonates in a highly deformed Triassic evaporite sequence. The single site in the Prebetic has not rotated significantly relative to stable Iberia.

In Malaguide rocks of the Sierra Espuña, in the adjacent Internal Zone, stable palaeomagnetic components were measured at one site in upper Miocene sedimentary rocks, at three sites in upper Oligocene–lower Miocene red marls, at one site in Oligocene marls, at one site in Jurassic limestones, and at two sites in Permo-Triassic red beds. The palaeomagnetic results suggest that about 60° of clockwise rotation occurred in the latest Oligocene–earliest Miocene, and a further 140° of clockwise rotation subsequently. One site in late Miocene sedimentary rocks yields unrotated declinations which, if not representing an overprinted direction, indicate that the rotation was complete by the end of the Miocene.

1. Introduction

The Betic–Rif arc of southern Spain and northern Morocco is a Late Cretaceous to Neogene mountain chain related to convergence between Africa and Eurasia. During the Neogene the older, dominantly metamorphic Internal Zones underwent extensional collapse, forming the Alboran Sea between Spain and Morocco and driving compressional deformation in the external fold and thrust belt [1]. The External Zones of the Betic Cordillera in Spain are divided into the Subbetic and Prebetic Zones (Fig. 1), consisting respectively of basinal and shelf-facies Mesozoic and Tertiary sediments that were folded and thrust northwestwards onto the Iberian platform during the Miocene [2,3].

Several workers have recently emphasized the importance of dextral strike-slip faulting in the Subbetic Zone [4–7]. De Smet [6], in particular, considered the whole Subbetic Zone to have been formed by distributed dextral shear centred on the Crevillente fault zone (Figs. 1 and 2). LeBlanc and Olivier [7], following Andrieux et al. [8], suggested that deformation in the External Zone was primarily a consequence of the westward motion of an Alboran microplate relative to both Iberia and Africa, and they suggested that the Internal/External Zone Boundary (IEZB) is a strike-slip contact, separating the Iberian margin from the Alboran microplate.

Recent palaeomagnetic studies in the Subbetic Zone have demonstrated a consistent pattern of clockwise rotations about vertical axes [9,10].

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These rotations confirm that there has been a component of dextral shear in the deformation, and Osete et al. [9] suggest that the rotational deformation occurred by distributed simple shear, either during Miocene oblique convergence, or during late Miocene–Recent deformation.

The purpose of this paper is to present the palaeomagnetic results of a detailed programme of investigation in the eastern Betic Cordillera, with the aim of determining the relationship between the vertical-axis rotations and the regional tectonics. In the first part we address the variation in rotational deformation across part of the External Zone, and its spatial relationship to thrusts and strike-slip faults, and in the second part we present evidence for the timing of rotation from unmetamorphosed sediments of the structurally highest tectonic unit of the Internal Zone (the Malaguide Complex).

2. Site selection and sampling strategy

Palaeomagnetic sites were required to show clear bedding, sedimentological or stratigraphic evidence for way-up, a relative coherence with the large-scale structure, and a minimum of internal deformation, and to cover a sufficient time interval to average out secular variation. Sites with gentle structural dips were preferred for sampling, although occasionally steeply inclined units were chosen where other suitable palaeomagnetic material was absent. Where possible,

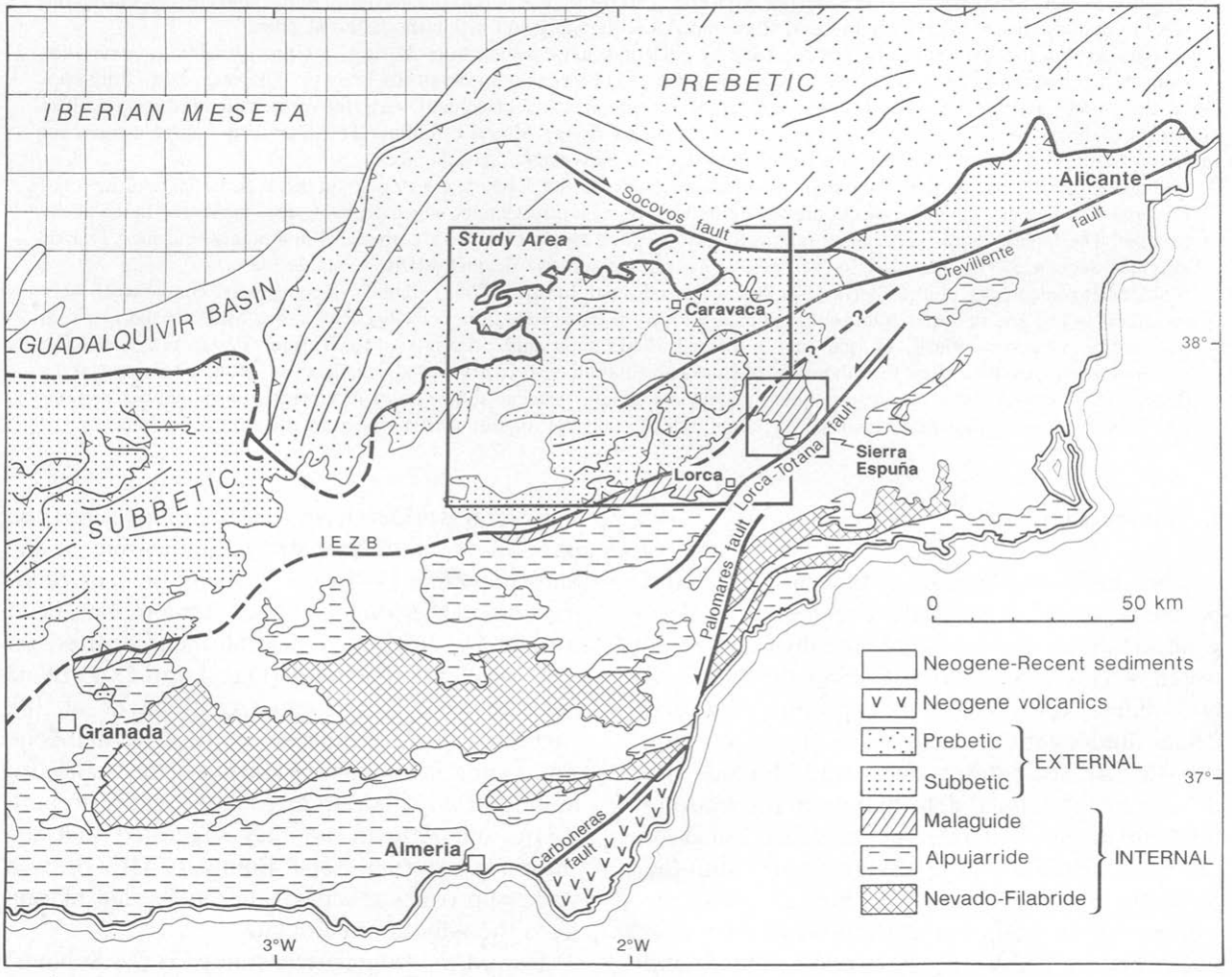


Fig. 1. Geological map of southeast Spain showing the location of the study area (Fig. 2), and the Sierra Espuña (Fig. 4). *IEZB* = Internal/External Zone Boundary.

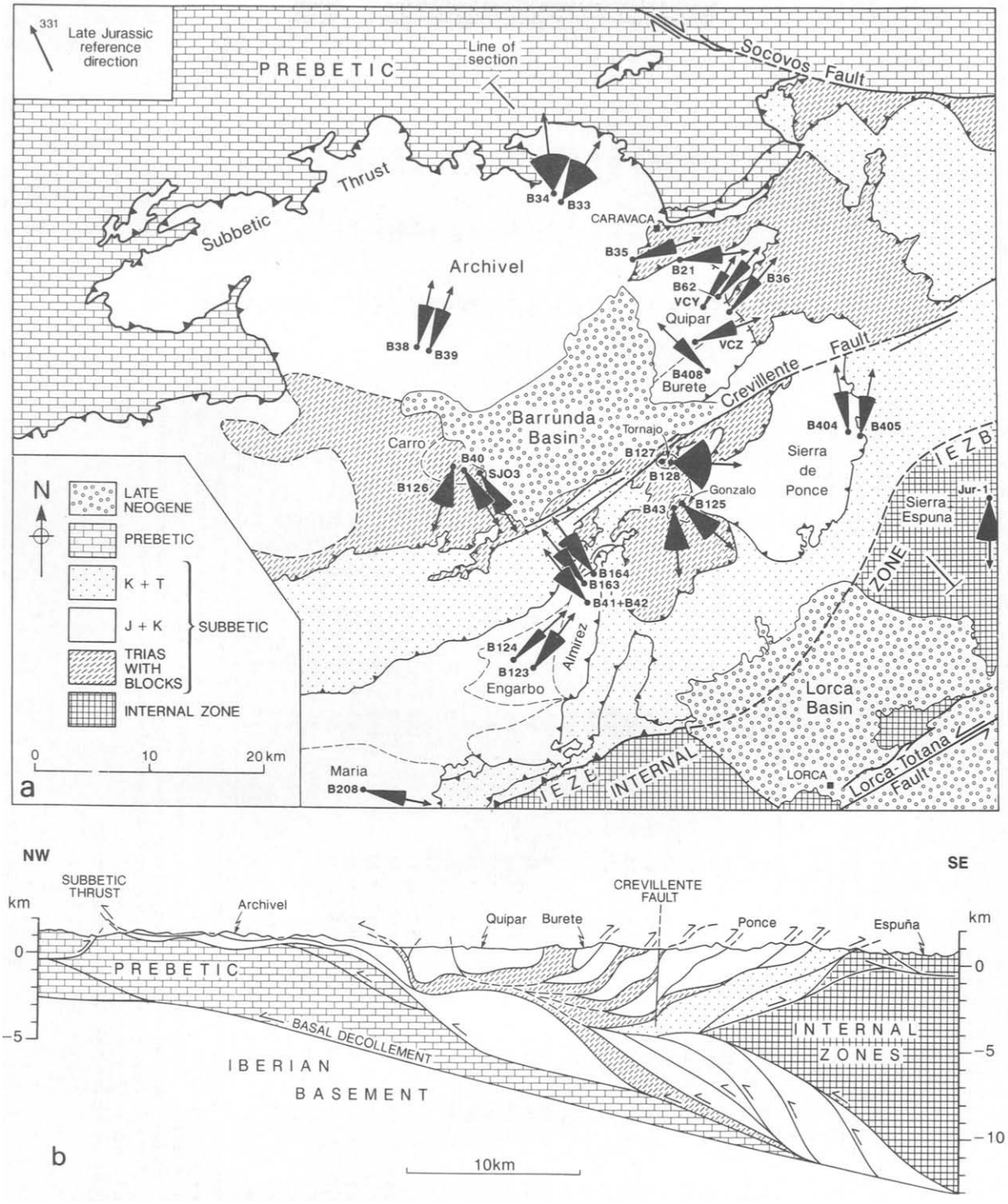


Fig. 2 (a) Simplified geological map of study area, showing Upper Jurassic mean site declinations with declination errors [36]. Also shown are results from the Upper Jurassic site in the Sierra Espuña and from two Lower Cretaceous sites in the Quipar block [12]. *IEZB* = Internal/External Zone Boundary. The line of the structural cross section (Fig. 2b) is also marked. The limits of exposure of structurally coherent blocks within the main thrust sheets are indicated by dashed lines. The nature of the block boundaries is often obscure. (b) Structural cross section through the External Zone of the Eastern Betic Cordillera, modified after [3]. Location shown in (a).

TABLE 1

Upper Jurassic (*Ammonitico rosso*) limestones, high-temperature components

Site	Before correction				After correction				N	n	n_r	Bedding		Type	Sheet	Grid Ref.		Block
	D (°)	I (°)	K	α_{95}	D (°)	I (°)	K	α_{95}				Dip	Direction			East	North	
B49	348	49	90.2	6.4	327	48	90.2	6.4	9	7	1	18	250	ii	888	7103	4104	Prebetic
B21	101	34	34.0	13.0	082	29	34.0	13.0	5	5	0	30	010	ii	911	7626	3907	Quipar
B36	062	46	50.8	7.3	046	50	50.8	7.3	9	9	0	15	310	ii	911	7654	3868	Quipar
B62	038	32	30.1	9.5	038	28	30.1	9.5	9	9	1	05	020	ii	911	7652	3837	Quipar
B33	045	81	44.6	9.1	034	51	24.4	12.5	7	7	2	30	030	ii	910	7493	3964	Archivel
B34	355	−01	11.5	20.6	354	29	11.5	20.6	8	6	2	30	180	ii	910	7496	3966	Archivel
B35 *	043	32	66.7	7.4	071	44	66.7	7.4	8	7	0	35	165	ii	910	7574	3906	Archivel
B37 **	017	70	34.8	9.5	038	72	34.8	9.5	8	8	0	06	130	ii	910	7447	3863	Archivel
B38	012	22	61.7	7.1	019	38	61.7	7.1	8	8	1	20	160	ii	910	7387	3830	Archivel
B39	148	−11	37.5	10.0	012	26	37.5	10.0	8	7	3	16	169	ii	910	7392	3828	Archivel
B40	154	−07	27.8	10.7	148	33	27.8	10.7	8	8	5	47	004	iii	931	7420	3726	Carro
B126	229	15	8.0	15.6	208	36	10.9	13.1	15	13	7	50	090	iii	931	7440	3718	Carro
SJ03	149	20	17.7	10.6	150	47	30.8	8.0	12	12	0	25	300	ii	931	7423	3722	Carro
B127 **	104	−18	51.0	9.5	111	04	61.6	8.6	8	6	0	52	354	iii	931	7598	3730	Tornajo
B128	108	35	7.7	29.5	090	33	7.7	29.5	6	5	1	25	014	iii	931	7603	3732	Tornajo
B43	177	09	12.2	16.5	174	23	12.2	16.5	9	8	2	17	027	lii	931	7612	3695	Gonzalo
B125	128	−40	21.0	11.5	132	25	12.7	15.0	12	9	8	60	320	iii	931	7606	3691	Gonzalo
B41 and B42 ***	306	31	16.9	13.9	319	51	17.1	13.8	12	10	2	28	010	ii	952	7532	3607	Almirez
B163	268	56	38.9	7.4	323	44	38.9	7.4	11	11	0	43	011	ii	931	7526	3623	Almirez
B164	291	61	19.3	12.9	323	30	19.3	12.9	8	8	5	41	353	ii	931	7530	3632	Almirez
B123	097	68	18.7	9.8	035	35	18.7	9.8	13	13	3	52	008	ii	952	7486	3556	Engarbo
B124	061	36	174.5	5.8	046	21	174.5	5.8	6	5	0	30	352	ii	952	7469	3567	Engarbo
B208	038	15	34.0	9.0	103	31	34.0	9.0	9	9	2	74	346	ii	952	7335	3427	Maria
B408	307	35	11.6	6.5	316	29	32.1	10.8	11	7	6	24	250	iii	911	7675	3825	Burete
B404	004	40	57.7	7.4	015	41	57.7	7.4	9	8	6	12	103	ii	932	7766	3755	Ponce
B405	337	−15	10.5	12.9	350	29	13.0	11.5	15	14	6	88	098	ii	932	7772	3743	Ponce

Directions are presented in normal polarity after bedding correction. D = declination; I = inclination; K = dispersion parameter; α_{95} = radius of 95% circle of confidence; N = total number of samples; n = number of samples in statistics; n_r = number of reversed polarity samples. Bedding is specified by dip and dip direction. 'Type' denotes sites with two (ii) or three (iii) component behaviour. Site locations are given as grid references from IGME Sheets 888, 910, 931–933, 952 and 954.

* B35 contains only two components and no reversals, and may be obscured by an overprinting intermediate-T component.

** Sites B37 and B127 have a high-T component which is almost completely obscured by an intermediate-T component and cannot be accurately resolved; it is not included in the discussion of the data.

*** B41 and B42 are two adjacent sites, which have been combined.

folds were sampled to test the relative age of the magnetisations.

Sampling involved drilling standard 2.5 cm diameter cores, oriented using both sun and magnetic compasses. Typically, between eight and fourteen cores were taken at each site, and between one and three samples were obtained from each core. Magnetisations were measured, in Oxford, on a CCL cryogenic magnetometer. Standard measurement procedure involved stepwise thermal magnetisation until no measurable signal could be identified (typically ~ 0.02 mA/m, or less than 2% of the initial intensity). Initially, bulk susceptibility was measured after each demagnetisation step to monitor growth of magnetic minerals during heating, but where no significant change was noted this was discontinued. Components were analysed on orthogonal plots using a least-squares routine. IRM acquisition and demagnetisation experiments were completed on one sample from each site to investigate the magnetic mineralogy.

3. Part 1: Mapping rotational deformation

Previous palaeomagnetic studies in the eastern Betic Cordillera for both magnetostratigraphic [11–15] and regional purposes [9,10] indicated that the Upper Jurassic pink–red nodular limestones of the *Ammonitico rosso* facies were good carriers of a palaeomagnetic signal. We completed a pilot study which suggested that, with the exception of some limited outcrops of Upper Cretaceous red marly limestones, little else was suitable for a regional palaeomagnetic investigation and so we have concentrated almost exclusively on the *Ammonitico rosso* facies. We also considered it important to restrict our sampling to a relatively narrow stratigraphic interval, to reduce uncertainties introduced by larger scale apparent polar wander.

Within the eastern Subbetic, Upper Jurassic the *Ammonitico rosso* facies consists of red–pink nodular biomicritic limestones with ages ranging from the Oxfordian through to the Tithonian [16]. The *Ammonitico rosso* facies has also been identified in one locality within the Prebetic (Site B49).

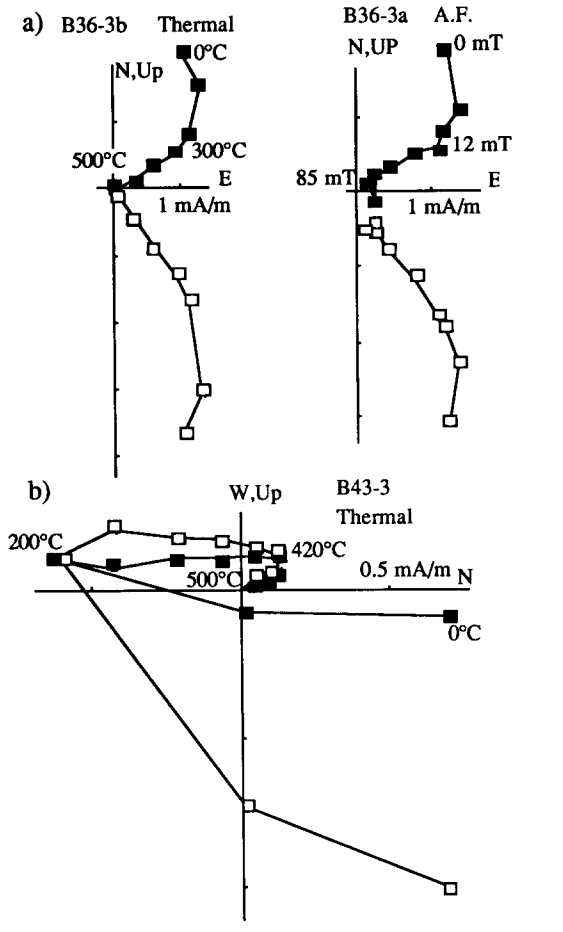


Fig. 3. Typical demagnetisation behaviour for Upper Jurassic *Ammonitico rosso* facies samples: Orthogonal vector diagrams (*in situ*; open symbols represent points in vertical plane; closed symbols, points in horizontal plane). (a) Example of two-component demagnetisation behaviour; Sample B36-3b (thermal) and B36-3a (AF). (b) Example of three-component demagnetisation behaviour; Sample B43-3 (thermal): A low-T component, corresponding to the present field, is removed by 200°C, an intermediate component is removed between 200 and 420°C, and a high-T component is removed between 420 and 500°C.

3.1 Remanence carriers and components

Demagnetisation treatment reveals two distinct types of behaviour. Most of the samples from the Subbetic *Ammonitico rosso* facies carry a low-temperature (T) component, unblocking below 260°C, and a high-T component, typically completely unblocked below 600°C (Fig. 3a and Table 1). AF demagnetisation produces similar results—low- and high-coercivity components

which can be equated with low- and high-T components isolated during thermal treatment. The high-coercivity component was not always completely removed at the highest field available in the AF system that we used (188 mT). Both normal and reversed directions are commonly observed within the high-T components from a single site (Table 1), and in some cases within a single sample, which may be a consequence of a reversal during the acquisition of the magnetisation. Similar behaviour has been documented in red limestones in Italy [17]. Site B49 from the Prebetic also exhibits two-component behaviour, on both thermal and AF treatment.

A few sites (seven) from the central part of the Subbetic display three-component behaviour—a low-T component, an intermediate-T component (260°C < T < 480°C), and a high-T component (480°C < T < 600°C) (Fig. 3b and Table 1). In a few cases the intermediate component is dominant and the high-T component very poorly defined. The intermediate component has a constant polarity at any single site, and generally has a direction distinctly different from that of the high-T component. AF treatment gives similar results, but the high-coercivity (high-temperature) component is typically poorly resolved.

At a few two-component sites we believe that the intermediate component completely obscures the high-T component. We identify these where a neighbouring site exhibits three components, and the intermediate component can be equated in direction with the higher T component in the two-component site (e.g., Sites B127 and B128). Samples from site B37 carry an intermediate component which is anomalously steep and does not decay to the origin. The high-T component at this site is not well defined, although the end point of the demagnetisation curve is relatively consistent.

The recognition of this overprinting component clearly throws some doubt on the nature of the high-T components at the sites that carry only two components. The intermediate component

TABLE 2

Fold tests

Fold	Before			After			N	K_{max}/K_{min}	Significance	McFadden [20]			Significance
	Dec.	*Inc.*	*K*	*Dec.*	*Inc.*	*K*				*CV*	Before	After	
True folds													
Ammonitico rosso: High-temperature component													
B38–B39	351	18	10.7	012	24	49.2	15	4.6	+ > 99%	Ambiguous			
B33–B34	003	48	3.3	012	40	6.7	13	2.0	+ > 95%	1.7	1.8	0.5	+ > 95%
B404–B405	346	06	5.2	358	34	13.7	22	2.6	+ > 99%	4.8	2.0	1.6	+ < 95%
B408	307	35	11.6	316	29	32.1	7	2.8	+ > 95%	2.1	2.7	0.9	+ > 95%
Sierra Espuña sites													
Aqu-2	157	29	2.8	129	59	9.4	15	3.4	+ > 95%	1.7	2.0	1.3	+ > 95%
Oli-1	207	42	18.9	219	28	9.5	26	2.0	− > 95%	3.5	4.3	8.2	− < 95%
PT-3	006	13	8.3	003	−6	22.0	15	2.65	+ > 95%	1.7	2.0	0.5	+ > 95%
Structurally coherent blocks													
Ammonitico rosso: High-temperature component													
Carro (B40,B126,SJ03)	179	14	3.6	172	43	7.7	33	2.1	+ > 95%	2.1	2.6	1.3	+ > 95%
Quipar (B21,B36,B62)	061	41	11.2	051	38	13.6	23	1.2	+ < 95%	2.1	0.7	0.1	+ < 95%
Almirez (B41,B42,B163,B164)	290	50	11.9	322	43	19.9	29	1.6	+ < 95%	2.1	2.3	1.7	+ > 95%
Engarbo (B123,B124)	080	60	11.9	039	31	20.7	18	1.7	+ < 95%	1.7	1.9	1.6	+ > 95%
Gonzalo (B43,B125)	154	−19	4.5	152	26	7.7	17	1.7	+ < 95%	Ambiguous			

Mean directions before and after application of a bedding correction. K = dispersion parameter. McElhinney [19] fold test parameter is K_{max}/K_{min}. McFadden test parameters: CV = critical value of test parameter at 95% confidence, and test parameters before and after tectonic correction. For McFadden [20], if test parameter > CV, there is a correlation between tectonic correction and magnetic direction, and the fold test is not significant. Result of the test is indicated as follows: negative (−) is magnetisation acquired after folding, and positive (+) is magnetisation acquired before folding. Significance is expressed as relative to a percent confidence factor; < 95% confidence is not considered significant. The McFadden test is considered ambiguous where the two test definitions give contradictory results. Maximum K at 26% unfolding.

only has a single polarity at any individual site, so two-component sites with a single-polarity high-T component may have been overprinted; these include sites B35, B36, B124 and B163. Of these, all but B35 can be directly linked to an adjacent site which has mixed-polarity high-T components (which can thus be considered reliable) in a similar direction, supporting the earlier age for the two-component remanence. Site B35 gives an inclination within the range expected for the Upper Jurassic, and samples that clean directly to the origin. We submit that it is unlikely that this site carries an overprinting magnetisation, although this cannot be completely discounted.

The high-T component passes each field test that we have been able to apply (see later for detailed discussion) and we believe it pre-dates the deformation. The intermediate-T component passes some fold tests and fails others so it is most likely to have been acquired during the deformation. The low-T component is of recent origin.

It is important to correctly identify the nature of the intermediate- and high-T components in two-component samples, so we have investigated the nature of the remanence carriers using orthogonal isothermal remanent magnetisation (IRM) acquisition and thermal demagnetisation of IRM [18]. The orthogonal IRM analysis indicates that low-, intermediate- and high-T components are carried by samples with variable proportions of both high- and low-coercivity material which probably corresponds to hematite and magnetite, respectively. The blocking temperature spectra for the two phases overlap: both are likely to contribute to the intermediate- and high-T components. The age of the formation of these components is probably best assessed by application of field tests.

3.2 Fold tests

Seven *Ammonitico rosso* sites were sampled on the limbs of a series of map-scale middle Miocene folds, giving four fold tests (see Table 2). These folds plunge less than 10°, and their axes are in a variety of orientations. All the fold tests show an increase in the dispersion parameter K for the high-T component. After unfolding all pass a McElhinny fold test [19] at 95% significance and two also pass a McFadden [20] fold test at 95% significance, indicating that the remanence pre-dates the folding. The B38–B39 fold gives an ambiguous result for the McFadden test because the fold is very tight. These results suggest that the high-T component pre-dates the middle Miocene folding episode.

In addition to these classic fold tests, palaeomagnetic directions from different sites within a single, coherent structural unit can be compared, before and after a simple strike-parallel bedding correction is applied. Although errors associated with folding about inclined axes may be included at this point, this approach can still provide a useful test of the nature of the components. We have applied this type of test to several groups of sites, listed in Table 2. The scatter of the high-T components is generally reduced after a tectonic correction has been applied, and the Carro (B40, B126, SJ03), Almirez (B41 and B42, B163, B164) and Engarbo (B124, B125) blocks show a statistically significant improvement after untilting, again suggesting that the magnetisation pre-dates middle Miocene folding.

The relative age of the intermediate-T component is less well defined. It is encountered within the true fold structure sampled at site B408, where it clearly fails the tests [19,20]. The intermediate component from the Don Gonzalo block (B43, B125) also fails a fold test, although the minimum test statistic occurs at 26% unfolding [20]. In contrast, the intermediate components from the Carro block (B40, B126) pass a McElhinny fold test at 95% significance, and those from the Tornajo block (B127, B128) pass a McFadden test. It seems probable that the intermediate components are the result of remagnetisation during the folding and thrusting event, possibly as a result of fluid migration and increased temperature and pressure [21].

3.3 Conglomerate test

A conglomerate with clasts of indurated pink–red limestone (Kimmeridgian–Tithonian *Ammonitico rosso* facies) and white, oolitic limestone (Dogger–Kimmeridgian) in a matrix of pink, micritic and calcarenitic limestone outcrops around the Piedra del Almirez, where it was sampled in a

single face of a quarry. The matrix of this conglomerate contains foraminifera that date the conglomerate as Eocene–lower Oligocene [Reicherter and Luterbacher, pers. commun.].

Analysis of the magnetisations revealed the presence of a low- and a high-blocking temperature component in each sample. From all samples, both clasts and matrix, the low-T component is significantly well clustered ($N = 37$, $R = 34.7$) about the present-field direction ($Dec. = 350°$, $Inc. = 61°$), suggesting that this component was acquired after the formation of the conglomerate, and probably represents a recently acquired magnetisation. The matrix preserves a significantly clustered high-T component ($N = 13$, $R = 10.7$), which is directed up, and to the southwest ($Dec. = 228°$, $Inc. = -19°$). As no bedding correction can be applied to this site, this direction is of limited regional significance. The high-T component has unblocking-temperature characteristics similar to those of the high-T components in the *in-situ Ammonitico rosso* sites, and is believed to be equivalent. The directions of the high-T component obtained from individual clasts are significantly well grouped (e.g., Clast 1, $N = 5$, $R = 4.9$), although the mean directions from all the clasts are statistically random ($N = 8$, $R = 3.1$); this component thus passes the conglomerate test, and pre-dates the formation of the conglomerate in the early Tertiary.

3.4 Discussion of results

The low-T component is consistently directed north and down, and fold tests indicate that it post-dates the major tectonic episode. It is probably a recent overprint which is not useful for tectonic analysis.

The directions of the intermediate components can be interpreted as due to remagnetisation during the folding and thrusting event. Similar evidence of remagnetisation has been discussed by Villalain et al. [22]. A multitude of rotational histories can explain these data, but we will not discuss them further.

The high-T component consistently passes fold tests, so the magnetisation pre-dates the Miocene folding event. The conglomerate test suggests that this component pre-dates the formation of an Eocene–early Oligocene conglomerate. Whilst this does not demonstrate that the magnetisation is primary, the relative consistency of the inclinations suggests that any early tilting of horizons before the acquistion of the magnetisation is minor. This high-T component should therefore prove a useful marker for tectonic rotations during the Tertiary deformational events.

The inclination values for the high-T components yield palaeolatitudes with a mean value of $20 \pm 8°$, which is consistent with the projected location of the southern margin of Iberia in the Late Jurassic (calculated [23,24] to lie between 18°N and 25°N).

The mean site declination data are displayed on Figure 2a. These show an interesting distribution: within any individual, coherent structural unit the results are consistent; yet between these units the declinations are highly variable. This suggests that the massive Jurassic limestone units deformed as relatively coherent blocks, with rotations accommodated by large displacements at their boundaries. The more marly Cretaceous and Tertiary sediments are more pervasively deformed. It is unfortunate that true fold tests were only available in the units which only show small rotations. It might be thought that the large rotations could be an artefact of simple tilt corrections being applied to unidentified plunging structures, in which case sites with steeper dip should show more anomalous rotation than gently dipping sites. However, this is not the case in our dataset. For example, Site B125 has a bedding dip of 60° towards 320° and a tilt-corrected high-T component of ($Dec. = 132°$, $Inc. = 25°$), while B43 from the same block has a dip of 17° towards 027° with a high-T component of ($Dec. = 174°$, $Inc. = 23°$). Similarly, B40 has a dip of 47° towards 004° with a high-T component of ($Dec. = 143°$, $Inc. = 33°$) while SJ03 for the same block has a dip of 25° towards 300° with a high-T component ($Dec. = 150°$, $Inc. = 47°$). There is no consistent relationship between rotation angle and bedding dip, giving us cause to be confident that our rotation estimates reflect real block rotations.

A wide variety of declinations are observed, from 311° through a majority between 030° and 060°, to 200°. This distribution is consistent with a general clockwise rotation from the reference di-

rection of 331°. This general result is consistent with that obtained by Osete et al. [9] and Platzman [10].

The distribution of rotational deformation is not symmetrical across the Subbetic. Four sites from the Archivel block (Sites B33, B34, B38, B39, Fig. 2) suggest that this thin but areally extensive thrust sheet [3] on the northern margin of the Subbetic has only rotated about 30° clockwise. The Quipar block (Sites B21, B36, B62) immediately to the south has rotated about 70° clockwise and this rotation also seems to have affected rocks at site B35 in the Archivel block adjacent to the thrust contact between the two. From the Quipar block south across the rest of the Subbetic rotations are highly variable, and in some cases very large. The largest ($> 130°$ clockwise) come from sites within the Carro, Tornajo and Don Gonzalo blocks, within the Median Subbetic [6]. These are in relatively small, isolated units of Jurassic carbonate shelf facies lying on an extensive, highly brecciated Triassic unit, with a large component of remobilised gypsum in the matrix. De Smet [6] has described this facies as part of the Crevillente strike-slip fault belt. The widespread distribution of this Triassic breccia, and its relationship to other units, however, suggests to us that it is a decollement carpet at the base of a major thrust sheet. Fragmentation of the relatively rigid Jurassic limestones and dispersal within this evaporatic breccia may have facilitated the extreme rotations and their proximity (< 10 km) to the dextral Crevillente fault zone may well have been a factor. Most of these large rotations are from sites that exhibit three-component behaviour (except for Site SJ03), and most sites that exhibit three-component behaviour have high-T components that indicate large rotations (except for Site B408). These exceptions give added confidence to the interpretation of the high-T components as indicating tectonic rotations, rather than being due to remagnetisation.

Two substantial blocks, the Sierra de Ponce (Sites B404 and B405), and the Sierra Burete (Site B408), in the central part of the Subbetic on either side of the Crevillente fault show little or no rotation, which suggests that this fault did not exert an over-riding control on the rotations. The northern part of the Sierra Almirez (Sites B41, B42, B163, B164) exhibits a small anticlockwise rotation of $10 \pm 8°$, in contrast to the dominant pattern of clockwise rotation.

The declination from Site B49 from the Prebetic Zone shows no significant rotation relative to the Iberian reference direction, suggesting that there has been little rotation of this part of the Prebetic Zone. The structure is far more coherent than in the Subbetic Zone, and the result from Site B49 suggests that the overall rotation in the Prebetic could be significantly less than in the Subbetic. However, some differential rotations about vertical axes may be expected to be associated with this highly arcuate fold belt.

3.5 Tectonic interpretation

The structure of the External Zones is dominated by fold and thrust structures, which are clearly shown in the published 1:50,000 geological maps of the region (Fig. 2). The Subbetic thrust in particular has a demonstrable displacement in excess of 20 km in a northwesterly direction [3,25], and kinematic data from the IEZB in the eastern Betic Cordillera show that it was a S- to SE-directed backthrust during the middle Miocene [25]. Folds and thrusts in the Prebetic and northern part of the Subbetic are dominantly WNW- to NW-directed, whereas in the central and southern Subbetic they are SE-directed. Where we have been able to define rotated blocks palaeomagnetically, these appear to correlate with structurally defined thrust blocks. Both large (e.g., Archivel) and small (e.g., Tornajo) rotated blocks lie on tectonic contacts that are gently dipping and are best described as thrusts.

The Subbetic is also affected by major strike-slip faults, but these appear to have formed in conjunction with the compressional deformation. The Serravallian–Langhian Socovos fault [26], which forms the Subbetic/Prebetic boundary in the northwest of Fig. 2a, passes northwest into the Prebetic, where it transfers displacement onto the thrusts in the western Prebetic (Fig. 1). Total displacement on the Socovos fault may be several tens of kilometres. The E-NE-trending Crevillente fault was probably active in a dextral sense in middle Miocene time, but was reactivated in a sinistral sense after the Messinian [27]. This fault dies out in the west of Fig. 2a, presumably transferring its displacement onto thrust faults or mi-

nor strike-slip faults. In view of this, it is unlikely to have a displacement of more than a few tens of kilometres.

Palaeomagnetically determined rotations are distributed widely across the eastern Subbetic Zone, and are not limited to the vicinity of the Crevillente fault zone. There is an overall trend to increasing rotation from north to south across the External Zone as a whole, although there is clearly much scatter about this trend. There is also a trend to larger rotations in smaller blocks, and some of the largest observed rotations come from small blocks in the south of the Subbetic.

These observations suggest that although the clockwise rotations must reflect a component of dextral shear in the deformation, the overall tectonic pattern is one of dextrally oblique convergence, taken up primarily by thrusting. The strike-slip faults, which are mainly located towards the internal margin of the thrust belt, probably reflect an element of deformation partitioning. The strike-slip faults cut, and must postdate, the thrusts in their vicinity, but they were coeval with thrusting in the more external parts of the system, which continued until latest Miocene time.

4. Part 2: Timing rotational deformation

To investigate the timing of tectonic rotation we have completed a palaeomagnetic study of a Permo-Triassic to upper Miocene sedimentary succession in the Sierra Espuña, part of the Malaguide unit of the Internal Zone of the Betic Cordillera.

4.1 Geological framework

The structure of the Sierra Espuña [28,29] is illustrated in Fig. 4. The Permo-Triassic to lower Eocene rocks of the Sierra Espuña were thrust into an imbricate stack in late Eocene times. A front to this thrust stack is preserved beneath a thick sequence of Oligocene conglomerates. Deformation propagated into the foreland basin during the late Oligocene with renewed thrusting. In early to middle Miocene time the whole stack was folded by the NW-vergent recumbent Espuña fold [28], and Subbetic rocks were thrust southeast over the Malaguide along the IEZB [25]. On

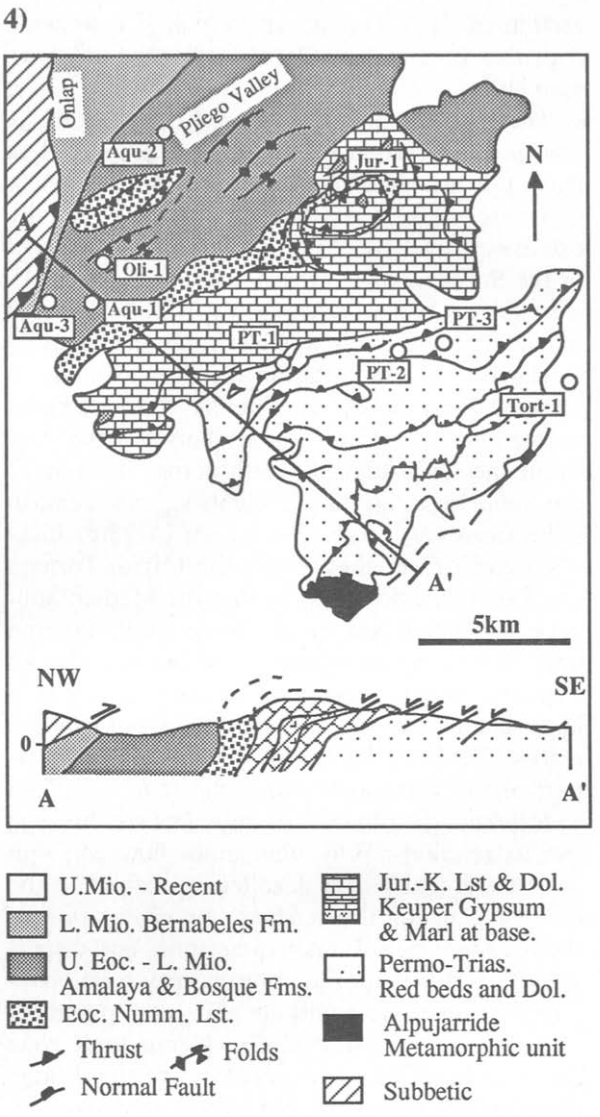


Fig. 4. Geological map of the Sierra Espuña with location of palaeomagnetic sampling sites and structural cross section [28].

the southern side of the Sierra Espuña Malaguide rocks are separated from underlying greenschist facies metasediments by a large NNW-directed low-angle shear zone that was active in the middle Miocene [29]. In the late Miocene, the thrust stack was dissected by normal faults. The Sierra Espuña as a whole is bounded to the southeast by the NE–SW-trending sinistral Lorca–Totana fault of late Miocene to Recent age, and to the southwest and northeast by the Neogene Lorca and Fortuna basins.

TABLE 3

Mean site directions of characteristic magnetisations: Sierra Espuña

Site	Before correction				After correction				N	n	n_r	Bedding		Grid Ref.		Iberian reference		Declination anomaly D (°)	
	D (°)	I (°)	K	α_{95}	D (°)	I (°)	K	α_{95}				Dip	Direction	East	North	D (°)	I (°)		
Tort-1	351	43	11.5	11.8	332	15	11.5	11.8	23	15	0	10	320	7957	3640	2	52	−11	$
Aqu-1	153	−4	31.9	22.2	151	31	32.5	22.0	9	3	3	40	340	7792	3668				
Aqu-2	157	29	2.8	27.9	129	59	9.4	13.2	17	15	0		Fold	7831	3728				
Aqu-3	141	6	25.2	12.2	142	41	23.7	12.7	22	7	7	36	316	7778	3669				
Aqu site means	150	10	18.8	29.3	142	44	25.6	24.9	3	3	2					2	52	140	
Oli-1	207	42	18.9	6.8	219	28	9.5	9.7	38	26	26		Fold	7795	3677	6	52	201	$
Jur-1	197	30	25.0	9.0	178	50	16.0	11.0	12	11	0	34	58	7882	3704	331	35	207	
PT-1	186	−6	27.0	12.0	185	6	31.0	12.0	8	6	6	30	290	7864	3644				
PT-2 *	228	22	4.0	29.5	228	51	4.2	28.5	9	9	1	47	42	7899	3650				
PT-3	186	−13	8.3	14.1	183	6	22.0	8.3	18	15	15		Fold	7912	3649				
PT site means	186	−9	268.1	15.3	184	6	3319.6	4.3	3	2	2					343	3	203	

Directions are presented in normal polarity after bedding correction. N = total number of samples; n = number of samples in statistics; n_r = number of reversed polarity samples (for site means, N = total number of sites, n = number included in statistics and n_r is number of sites); α_{95} = radius of 95% circle of confidence; K = dispersion parameter. Bedding is specified by dip and dip direction. Where a site is taken over a fold no bedding is given. Site locations are given as grid references from IGME Sheets 932, 933 and 954. The declination anomaly is calculated from the Iberian reference direction [23,24] with a correction for the rotation of Iberia in the Late Cretaceous [39] and the mean declination after tectonic correction.

* Site PT-2 is not included in the site mean (see text for discussion).

$ For these sites the declination anomaly is calculated from the before correction declination.

4.2 *Upper Miocene*

Upper Miocene (Tortonian) marine marls from the Neogene extensional basin on the eastern edge of the Sierra Espuña were sampled at one site in grey argillaceous beds. NRM intensities are relatively high, up to 9 mA/m. The magnetic mineralogy is dominated by a high-coercivity, high-T (650°C) phase, presumably specular haematite [30]. The coarser grained clastic Tortonian detritus was derived from the Permo-Triassic red beds and dolomites that are now exposed in the Sierra Espuña; it is likely that red beds were also the origin of the haematite seen in the fine-grained rocks.

Thermal demagnetisation reveals clear, stable, north-down directed components (Table 3 and Fig. 5a). As these rocks are not folded, no fold test can be applied, so it is difficult to assess the age of the magnetisation. The inclinations before any tilt correction is applied are only slightly gentle ($43 \pm 12°$) compared to those predicted for the late Miocene (reference $Inc. = 52°$). The sequence dips gently ($\sim 10°$) to the northwest, away from the sediment source in the Sierra Espuña. Application of this correction reduces the inclina-

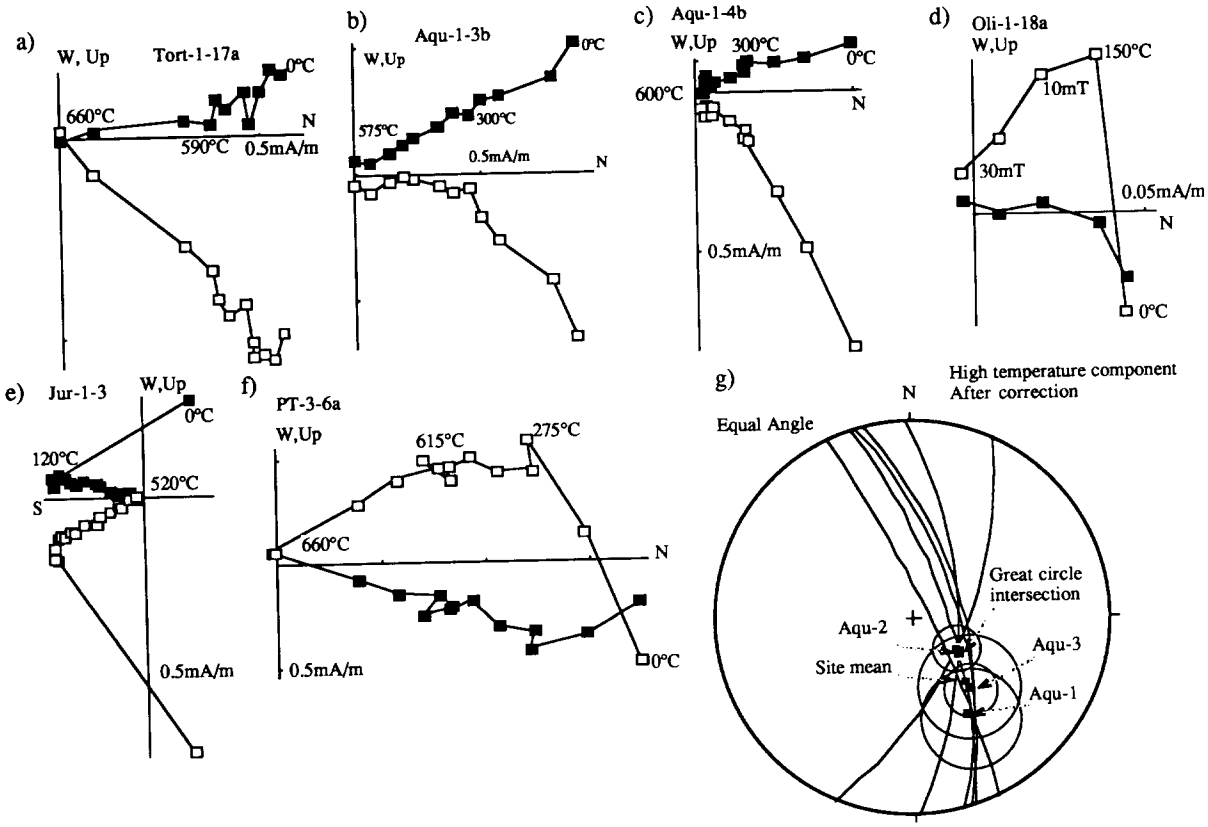


Fig. 5. Examples of thermal demagnetisation in orthogonal vector projection (open symbols represent points in the vertical plane; closed symbols in the horizontal plane; *in situ*). (a) Upper Miocene (Tortonian) marl. (b) Upper Oligocene–lower Miocene (Aquitanian) marl (Amalaya Formation): Example of two components isolated by thermal demagnetisation. (c) As (b). Example of a complete overprint. (d) Oligocene marls (Bosque Formation): Examples of thermal demagnetisation to 150°C followed by AF demagnetisation. (e) Upper Jurassic limestone. (f) Permo-Triassic red beds. (g) Upper Oligocene–lower Miocene (Aquitanian) marls: Best-fit great circles through picked components from samples with a common bedding orientation, and mean site directions on a lower hemisphere equal-area stereographic projection after a structural correction has been applied. The great circle intersection corresponds to the high-T component site mean directions.

tion still further (corrected *Inc.* = 34°). It is not clear that the measured bedding represents a true palaeohorizontal within this coarse clastic sequence, and a dipping depositional surface may be being represented.

Tauxe et al. [31] have used anisotropy of magnetic susceptibility (AMS) to demonstrate an early remanence in sediments exhibiting a depositional magnetic fabric. These Tortonian sediments have a typical sedimentary AMS fabric—oblate, with the minimum axis subvertical. The directions of the maximum axes are consistent, with the mean directed towards the northeast, perhaps reflecting a dominant current flow direction. The oblate fabric is probably largely a consequence of compaction. We suggest that the possible presence of a sedimentary fabric is an argument for an early, pre-compaction remanence in these sediments. The mean direction before correction (*Dec.* = 351°, *Inc.* = 43°, $\alpha_{95} = 12°$) is within error of the Miocene reference direction, so if this component is primary there has been no rotation since the late Miocene.

4.3 Upper Oligocene–lower Miocene

Red, yellow and grey marls interbedded with coarse sandstone and polymict conglomerates of the Oligocene to Aquitanian Amalaya Formation were sampled from three sites. The magnetisation is dominantly carried by a high-coercivity phase (probably haematite) which typically unblocks by about 650°C. NRM values are in the range 0.25–2.5 mA/m. A high-T component exists in most samples (e.g., Fig. 5b), although it is often difficult to define, being partially or completely overprinted by a present-field component (Fig. 5c).

The fold test (Table 3, Aqu-2) is positive, and significant at the 95% confidence level, indicating that the magnetisation pre-dates the formation of the fold, despite the poor resolution of the direction after the bedding correction has been applied.

A great-circle analysis [32] (Fig. 5g) was applied to the picked directions from all the samples, including those which are clearly overprinted by a present-field component, with a single circle being defined from all the samples with a common bedding orientation. When the intersection of these circles, after bedding correction, are compared to the mean site directions of the high-T component, a reasonable correlation is noted, as it falls within the confidence circle of the mean directions, suggesting that the demagnetisation technique was largely successful in defining the characteristic high-T component in these samples.

The mean direction obtained from the upper Oligocene–lower Miocene has a declination of 142° and an inclination of 44° ($\alpha_{95} = 26°$, Table 3). The inclination is within error of the reference inclination and the difference between the reference declination and the mean site declination indicates a 140° clockwise rotation.

4.4 Oligocene

Marls of the Oligocene Bosque Formation, sampled at a site across a minor (wavelength = 2–3 m) anticline, gave stable components. The initial intensities were relatively weak, varying from 0.1 to 0.3 mA/m. Thermal treatment removed a present-field component by about 120°C, and then started to isolate a higher T component. In the majority of cases this component was swamped by magnetisations carried by newly grown minerals, indicated by an increase in the bulk susceptibility. AF treatment quickly removed a low-coercivity component, leaving a high-coercivity, present-field component.

The low-T–high-coercivity component is probably carried by goethite, and the high-T–low-coercivity component by magnetite. Other high-coercivity–high-T components appear on thermal demagnetisation of the IRM, but some of these may be grown during the thermal treatment, and may not have contributed to the NRM. As there is some overlap of the coercivities of magnetite and goethite, AF treatment on its own is not very successful in isolating the 'magnetite' component, so a compromise treatment procedure was devised to avoid these problems: a low-T step of 150°C removed the low-T component; subsequent AF treatment then isolated a consistent signal, even with very weakly magnetised samples (Fig. 5d).

The fold test is significantly negative (at 95% confidence), suggesting that the magnetisation was acquired after folding (Table 3, Oli-1). The

in-situ direction (Table 3) gives an inclination (42°) consistent with its predicted palaeolatitude, a declination of 207° (normal polarity), and a clockwise rotation of 201°.

4.5 Upper Jurassic

The Upper Jurassic is represented by yellowish/pinkish, faintly nodular, fine-grained limestone [16,33]. Only one site (Jur-1) within this stratigraphic unit has yielded stable high-T components. Samples from the two other sites sampled are dominated by low-T components, unblocking by about 120°C.

The samples from Jur-1 are more strongly magnetised than those from the other two sites, with NRM values between 0.75 and 0.2 mA/m. All samples from this site show similar demagnetisation behaviour: a present-field component (presumably carried by goethite) is removed by about 120°C, and then a stable, high-T component, probably carried by magnetite, is progressively removed, completely unblocking at about 520°C (Fig. 5e).

The age of the stable component from site Jur-1 has not been assessed by appropriate field tests as none have been identified. Other Jurassic limestones in the Subbetic, of approximately the same age but of different lithology, host magnetisations that pre-date a Miocene folding event.

Tectonic correction does not significantly change the direction of the high-T component, which is south and down (*Dec.* = 178°, *Inc.* = 50°, α_{95} = 11°, Table 3). This does not fit with reference directions from the Jurassic or younger successions, so this site must have been subjected to approximately 150° anticlockwise or 210° clockwise rotation. The inclinations from this site are rather steep, but within error of those predicted for Iberian sites in the Late Jurassic.

4.6 Permo-Triassic

Three sites were sampled in the upper thrust imbricates of the thrust stack in medium-grained, quartz-rich, red sandstones. These red bed facies have a magnetic mineralogy dominated by a high-T, high-coercivity component (probably specular haematite [30], with initial intensities between 2.0 and 0.3 mA/m. Thermal demagnetisation revealed a low-T, approximately present-field component, stable to about 250°C, and a high-T component, which is removed by about 650°C (Fig. 5f and Table 3).

One of the sites (PT-3) spanned a minor anticline, enabling a fold test which proved positive for the high-T component. This magnetisation (*Dec.* = 185°, *Inc.* = 06°, α_{95} = 22°) thus pre-dates folding, which probably formed during Eocene thrusting. The gentle inclinations are consistent with the equatorial palaeolatitude of Iberia predicted for the Permian [23,24]. This perhaps gives greater confidence in the age of the magnetisation, although it is recognised that a positive fold test does not necessarily indicate that the inclinations are correct, as the magnetisation could have been acquired when the beds were planar, but tilted.

One site (PT-2) yields a southwesterly, down-directed component. These components are frequently less well defined and the distribution of stable components from this site is significantly different to that of the others; the direction is not included in a calculation of the mean for this lithology.

Makel et al. [34] have published palaeomagnetic results from two Permo-Triassic sites in the Sierra Espuña, which they compare with European and African apparent polar wander paths, to assess the palaeogeographic position of the Malaguide Complex. Within the present context of local block rotations this approach is not considered appropriate. One of the sites that Makel et al. have presented (ESP I) is from a quarry in the southern part of the Malaguide Complex exposed in the Sierra Espuña, which has been mapped [28] as within an isoclinally folded sequence. Directions from this site are likely to be unreliable. The second site (ESP II) is from a location close to PT-3, and yields a direction (*Dec.* = 336°, *Inc.* = 01°, α_{95} = 14°), just within error of the mean site direction described here, although the difference in the directions may reflect differential rotational deformation between individual thrust sheets.

The amount of rotation recorded in these Permo-Triassic rocks is ambiguous, because where inclinations are subhorizontal it is difficult to distinguish between a normal polarity, unrotated direction, and a reversed polarity, rotated direc-

tion. If the mean directions are reversed (as is most probable for Permian times), the north and slightly up (or south and down, if the results are projected to a normal polarity) direction requires a large rotation to have affected these sites, consistent with the results from the Oligocene and Jurassic sites. If, however, the mean directions are assumed to be normal, the inclination error still overlaps that of the Permian reference direction, so it is possible that the true mean is unrotated.

4.7 Discussion of results and timing of rotation

All sites from the Permo-Triassic to Oligocene that show a stable magnetic remanence are systematically discordant with reference directions for stable Iberia, if it is accepted that the Permo-Triassic sites contain a reversed remanence. There is a remarkable consistency between sites over this time span in that they all lie between 200° and 207° clockwise of the stable reference direction for the equivalent age (Fig. 6 and Table 3).

Sites in the Oligocene–Aquitanian Amalaya Formation yield a clockwise rotation of 140°, which is less than the rotation observed in the older rocks. This suggests firstly that the rotations in the older rocks are clockwise, rather than anticlockwise, and secondly that about 60° of the 200° clockwise rotation in the older rocks took place between the magnetisation of the Oligocene Bosque Formation (which post-dates folding) and the magnetisation of the Amalaya Formation. The rotation may therefore have taken place during deposition of the Amalaya Formation.

The lack of rotation in the upper Miocene (Tortonian) suggests that the rotation was complete by the Tortonian if the magnetisation is primary.

4.8 Tectonic implications

The palaeomagnetic results suggest that the Sierra Espuña rotated about 200° clockwise as a discrete coherent block during the early to middle Miocene. The rotation may therefore have occurred during convergence between the Sierra Espuña and the Subbetic, accommodated along the gently NW-dipping backthrust that defines the IEZB in this area. Rotation may also have

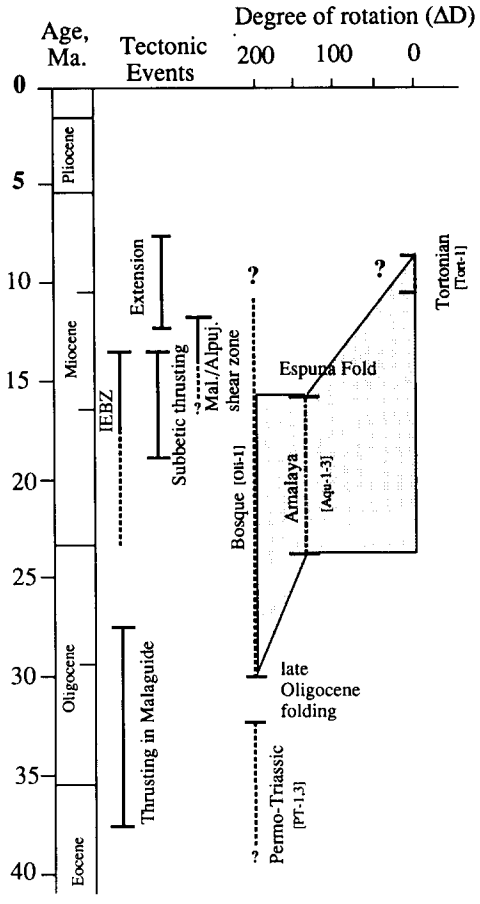


Fig. 6. Relative timing of tectonic events and palaeomagnetic rotation. The constraints on the relative ages of the magnetisations from the units sampled are shown against the degree of clockwise rotation. The shaded region signifies the interval in which the rotation could have occurred; the actual rotation path falls within this region. Age scale after [38]; timing of tectonic events after [28].

been accommodated on the south side of the Sierra Espuña along a low-angle NNW-directed shear zone separating Malaguide rocks from the underlying greenschist facies rocks of the Alpujarride Complex [29]. This was probably active in the middle Miocene, and may be a normal-sense shear zone comparable to other extensional structures of early to middle Miocene age described in the Internal Zones [1,35]. The other boundaries of the Sierra Espuña are overlapped unconformably by Neogene to Recent sediments, and the faults defining the block may be buried under the Neogene basins or may even have been reactivated as bounding faults to the intramontane

basins. One plausible way of explaining all the data is that the rotation occurred gradually as the basinal deposits of Oligo-Miocene Amalaya and Bernabeles Formations were being deposited, while the Espuña fold was forming. This allows for the lesser rotation in the Oligo-Aquitanian and for the top of the Bernabeles Formation to onlap onto the Subbetic where the point of onlap either marks the edge of the block or signifies the end of the rotation. Basinal sedimentation synchronous with the rotation would fill any 'holes' that must be a consequence of large magnitude block rotations.

The maximum time interval (latest Oligocene–Recent) available for the ~200° of rotation is about 30 Ma, giving a rotation rate of about 7°/Ma. Structural arguments suggest that the rotation may have finished earlier, in late Langhian times, as sediments of this age onlap, and are not cut by the IEZB; it is possible, however, that displacement of the Internal Zone relative to the External Zone continued on another thrust within the External Zone.

The main boundaries between the tectonic blocks in the Subbetic and the IEZB itself are all gently dipping thrusts, and rotation appears to have been accommodated by these faults. Thus lower–middle Miocene deformation at the IEZB and in the External Zone is probably a result of convergence oblique to the Iberian margin, rather than strike-slip movement parallel to the margin.

5. Conclusions

An early (pre-Eocene–early Oligocene) magnetisation can be isolated in Upper Jurassic *Ammonitico rosso* facies limestones of the external Betic Cordillera. Magnetic inclinations in these rocks yield palaeolatitudes consistent with the position of Iberia in the Late Jurassic. Declinations in the Subbetic Zone show highly variable clockwise rotations and sites from geologically defined thrust sheets exhibit similar amounts of rotation. They show no spatial association with strike-slip faults. One site from the Prebetic Zone is unrotated.

The Malaguide rocks of the Sierra Espuña in the Internal Zone have undergone a clockwise tectonic rotation of about 200° since the latest Oligocene. If the upper Miocene rocks in the area record a contemporary magnetisation, the rotation was complete before the Tortonian. A partial clockwise rotation of about 140° is recorded in upper Oligocene–lower Miocene sediments, suggesting that rotation may have started during Oligo-Miocene basinal sedimentation in the Sierra Espuña. The SE-directed thrust along the IEZB probably acted as a boundary to the rotating Espuña block. If so, rotation ceased at the same time as slip on this boundary, during middle Miocene times. Using the time limits discussed for the rotation, a minimum estimate of the rotation rate is about 7°/Ma.

The rotation of the Sierra Espuña overlaps the start of major thrusting in the Subbetic Zone. It is likely that rotational deformation also affected the Subbetic at this time.

Rotations about vertical axes are an integral part of the Miocene compressional deformation in the eastern Betics, and are probably the result of oblique convergence between the Internal Zone and the Iberian margin.

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